Measurement of Growth and Development

Growth Chart



Third Stage
Dr. Moustafa Alshammari
Growth and Developments

Learning objectives

- Define a growth chart and explain why a pattern over time matters more than a single point.
- Select the correct chart by age and sex (boys and girls; WHO and CDC).
- Explain the difference between a growth standard (WHO) and a growth reference (CDC).
- Measure weight, length/stature and head circumference accurately.
- Plot, read and interpret percentiles using the correct cut-offs.
- Apply the nursing process: act on findings, document, educate parents, refer.

Growth Charts

* Growth charts use a series of percentile curves to demonstrate the distribution of body measurements in children.

Growth Charts

3. Children who show a sudden increase (except during puberty), decrease, or no change in a previously steady growth pattern

* Because growth is a continuous but uneven process, the most reliable evaluation lies in comparing growth measurements over time. It is important to remember that normal growth patterns vary among children the same age

WHO Standards Monitor Infants Under 2 Years While CDC Charts Guide Ages 2 to 20

Assessment Feature	WHO Growth Standards (0 – 2 Years)	CDC Growth Reference (2 – 20 Years)
Target Population	Infants & toddlers from birth to 24 months.	Children and adolescents aged 2 to 20 years.
Biological Standard	Depicts how healthy children SHOULD grow under optimal environmental & feeding conditions.	Depicts how children ACTUALLY grew in the US during reference periods.
Feeding Baseline	Based on predominantly breastfed infants for at least 4 months (optimal feeding standard).	Includes mixed formula and breastfed infants from historical US health surveys.
Clinical Focus	Growth velocity, length-for-age, weight-for-length, and head circumference.	Stature-for-age, weight-for-age, and Body Mass Index (BMI)-for-age.

Growth Percentiles Establish a Child's Relative Position within Reference Norms

Understanding Growth Percentiles

Percentile Definition: A statistical value indicating the percentage of reference children who fall at or below a specific measurement.

50th Percentile: Represents the exact median parameter for age and sex. Half of healthy peers are larger, half are smaller.

Normal Range: Generally defined between the 5th and 95th percentiles on standardized WHO/CDC curves.

Trend vs. Static Value: A single percentile reading is merely a snapshot. Serial measurements over time (the growth curve) are clinically vital.

Case Scenario: 10th Percentile Height

If a 12-month-old boy is in the 10th percentile for length:

- He is taller than 10% of 12-month-old boys.
- 90% of peers are taller than him.
- Clinical Assessment: HEALTHY if he has consistently tracked along the 10th curve since birth.

"Percentiles are not grades in school. Being in the 10th percentile is not 'worse' than being in the 90th percentile. Consistency along an established trajectory is the primary indicator of normal physical growth."



* These children of identical age (8 years) are markedly different in size. Child on left, of Asian descent, is at 5th percentile for height and weight. Child on right is above 95th percentile for height and weight. However, both children demonstrate normal growth patterns.

Steps 2–6: select, record, calculate, plot, interpret

2 Select the chart

By age and sex; WHO <2 y, CDC ≥2 y.

3 Record data

Parents' stature, gestational age and birth data (<2 y); date, age, weight and length/stature immediately after measuring; comments.

4 Calculate BMI (≥2 y)

BMI = kg ÷ cm ÷ cm × 10,000, one decimal place.

5 Plot

Age on the horizontal axis; draw a vertical line up. Measurement on the vertical axis; draw across. Dot at the intersection. (Weight-for-length: length on the horizontal axis.)

6 Interpret

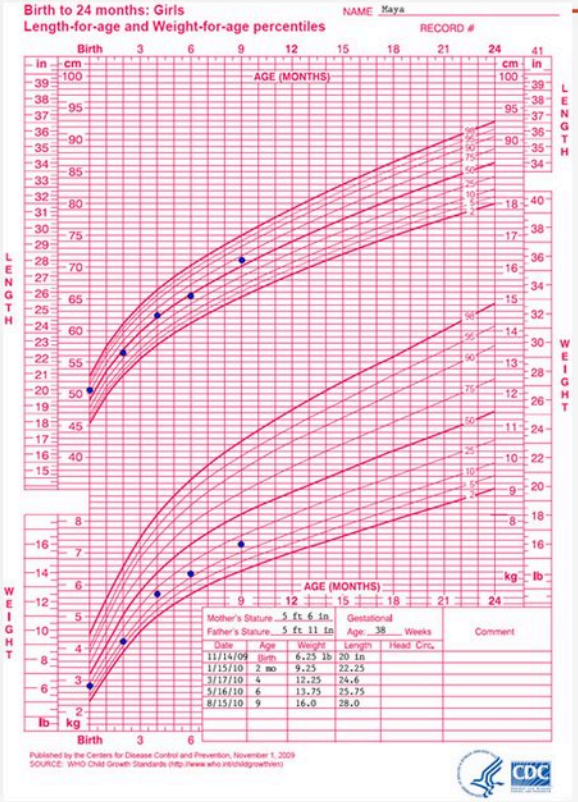
Read the percentile; apply the cut-off for the index; compare with previous visits for major shifts.

Age rounding

<2 y: nearest month. 2–20 y: nearest ¼ year.

Source: CDC (2013) Use and Interpretation of the WHO and CDC Growth Charts, Steps 2–6.

Worked example: plotting a girl's growth (WHO chart, 0–24 months)



Age	Weight (as charted)	Weight (kg)	Length (as charted)	Length (cm)
Birth	6.25 lb	2.84	20 in	50.8
2 months	9.25 lb	4.20	22.25 in	56.5
4 months	12.25 lb	5.56	24.6 in	62.5
6 months	13.75 lb	6.24	25.75 in	65.4
9 months	16 lb	7.26	28 in	71.1

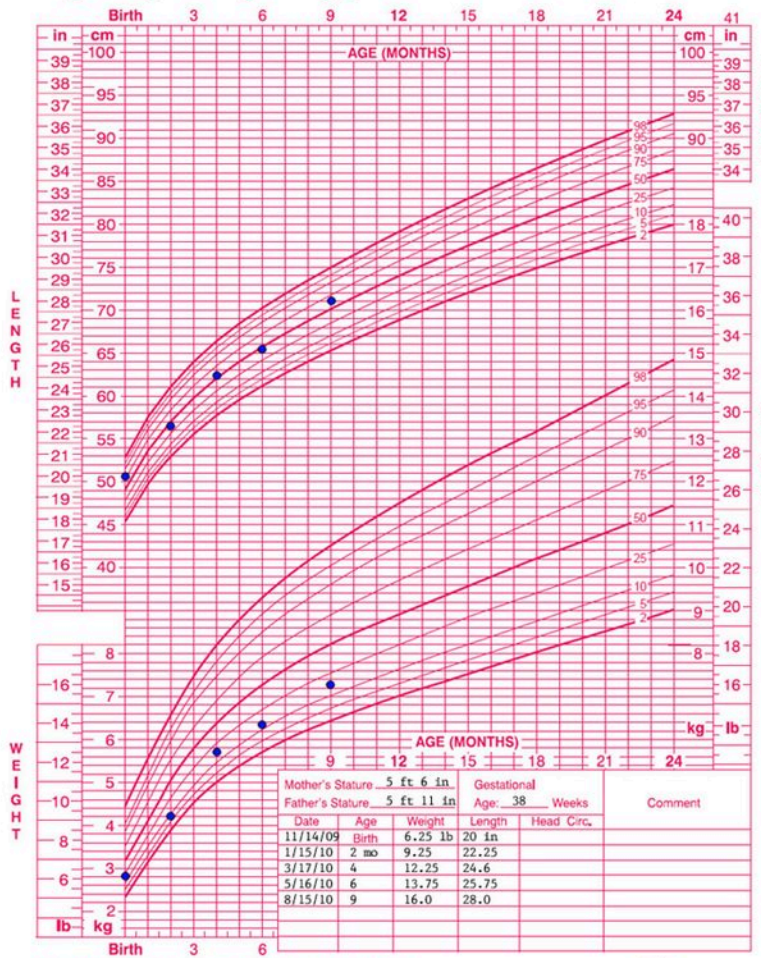
- Approximate reading: length points lie on or above the 50th curve; weight points lie below the 50th curve.
- Each series follows a similar curve over time (tracking).
- Next: plot weight-for-length and review feeding before any conclusion.

Chart image: CDC/WHO Birth to 24 months, Girls (CDC training example). Conversions: 1 lb = 0.4536 kg; 1 in = 2.54 cm.

Birth to 24 months: Girls
Length-for-age and Weight-for-age percentiles

NAME Maya

RECORD #



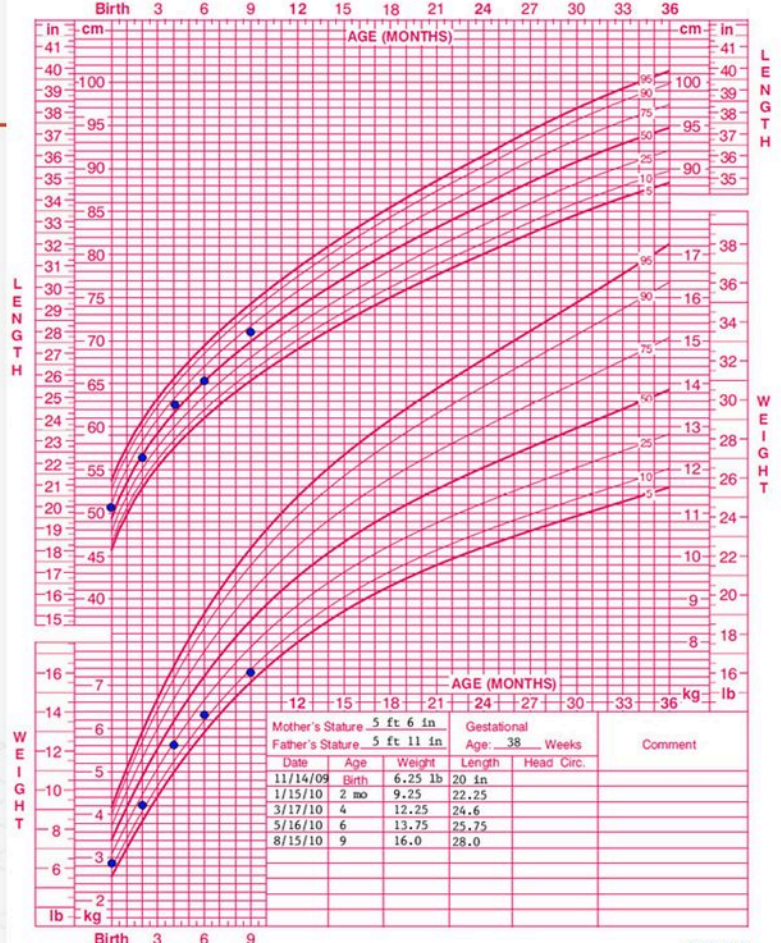
Mother's Stature 5 ft 6 in			Gestational		
Father's Stature 5 ft 11 in			Age: 38 Weeks		Comment
Date	Age	Weight	Length	Head Circ.	
11/14/09	Birth	6.25 lb	20 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

Published by the Centers for Disease Control and Prevention, November 1, 2009
SOURCE: WHO Child Growth Standards (http://www.who.int/childgrowth/en)

Birth to 36 months: Girls
Length-for-age and Weight-for-age percentiles

NAME Maya

RECORD #



Mother's Stature 5 ft 6 in			Gestational		
Father's Stature 5 ft 11 in			Age: 38 Weeks		Comment
Date	Age	Weight	Length	Head Circ.	
11/14/09	Birth	6.25 lb	20 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

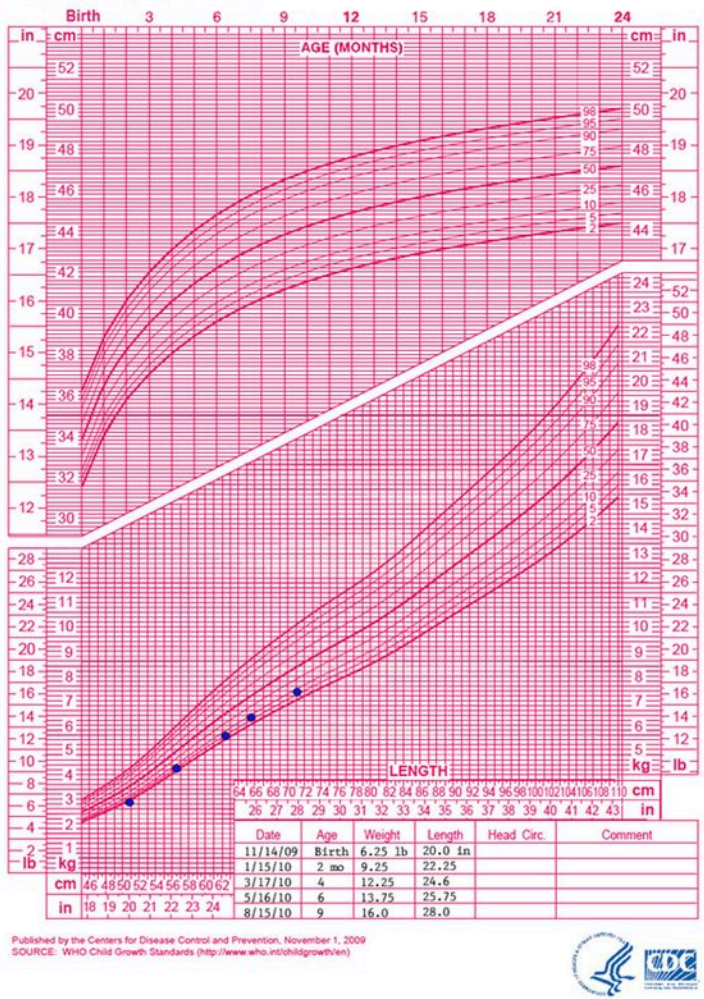
Published May 30, 2000 (modified 4/20/01).
SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
http://www.cdc.gov/growthcharts

SAFER · HEALTHIER · PEOPLE™

Birth to 24 months: Girls
Head circumference-for-age and Weight-for-length percentiles

NAME Maya

RECORD # _____



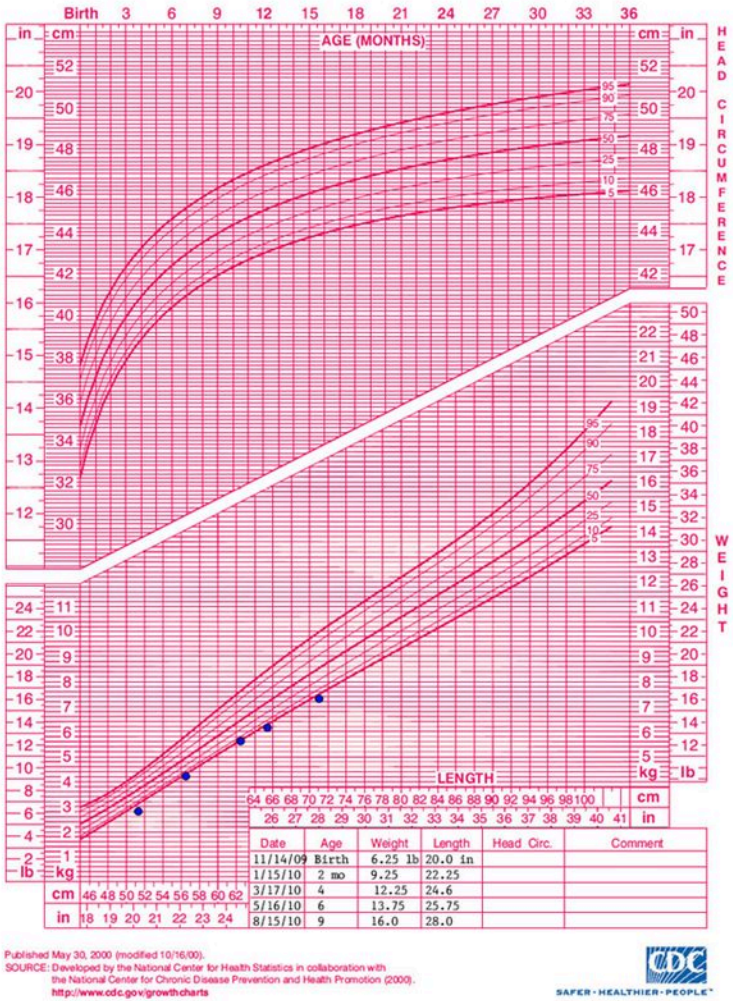
Date	Age	Weight	Length	Head Circ.	Comment
11/14/09	Birth	6.25 lb	20.0 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

Published by the Centers for Disease Control and Prevention, November 1, 2009
SOURCE: WHO Child Growth Standards (http://www.who.int/childgrowth/en)

Birth to 36 months: Girls
Head circumference-for-age and Weight-for-length percentiles

NAME Maya

RECORD # _____



Date	Age	Weight	Length	Head Circ.	Comment
11/14/09	Birth	6.25 lb	20.0 in		
1/15/10	2 mo	9.25	22.25		
3/17/10	4	12.25	24.6		
5/16/10	6	13.75	25.75		
8/15/10	9	16.0	28.0		

Published May 30, 2000 (modified 10/16/00).
SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
http://www.cdc.gov/growthcharts

SAFER · HEALTHIER · PEOPLE™

Interpreting the plotted point: cut-offs

Chart set	Index	Percentile	Nutritional indicator
WHO (<2 y)	Length-for-age	< 2nd	Short stature
WHO (<2 y)	Weight-for-length	< 2nd	Low weight-for-length
WHO (<2 y)	Weight-for-length	> 98th	High weight-for-length
CDC (2–19 y)	BMI-for-age	< 5th	Underweight
CDC (2–19 y)	BMI-for-age	5th to < 85th	Healthy weight
CDC (2–19 y)	BMI-for-age	85th to < 95th	Overweight
CDC (2–19 y)	BMI-for-age	≥ 95th	Obesity
CDC (2–19 y)	Stature-for-age	< 5th	Short stature

Source: CDC (2013) guide, Step 6 table. Healthy-weight band (5th to <85th) is the CDC BMI-for-age category. Percentile ranks signal need for monitoring, not diagnosis.

Growth patterns that need attention

- Height and weight percentiles widely disparate (for example, height near the 10th and weight near the 90th).
- Failure to follow the expected growth velocity, especially during rapid growth in infancy and adolescence.
- A sudden increase (except in puberty), decrease or no change in a previously steady pattern.
- A major shift in percentile rank compared with previous visits (CDC).
- A plotted point beyond the cut-offs for that index (previous slide).

Before escalating, check

Equipment calibration and technique · correct chart and sex · age (corrected for preterm) · units · parents' stature, birth weight, gestational age · chronic illness or special health care needs.

Sources: CDC (2013) guide; first three bullets carried over from the original course deck (textbook-based) — verify wording against the textbook page.

Nursing procedure in health institutions (nursing process)

Phase	Nursing actions
Assessment	Prepare and check equipment; identify the child; explain to the caregiver; measure; collect gestational age, birth weight, feeding history, parents' stature, illness; select chart; plot; compare with previous plots.
Diagnosis	Identify actual or risk problems related to growth and nutrition using the institution's taxonomy.
Planning	Set measurable goals with the family; set the follow-up measurement schedule.
Implementation	Feeding and breastfeeding support; nutrition counselling; parent education; report or refer to the physician or dietitian when a cut-off or pattern concern is present; document.
Evaluation	Re-measure at follow-up; compare the new plot with earlier points; revise the plan.

Thank you